

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 4: Germination: Process, Types and Conditions Necessary for Germination in Maize and Bean Seeds

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	A	D	D	D	A	C

USE AFTER: PREDICT PHASE — Group prediction chart for the 5 set-ups: the variable to change and the ones to keep the same

Q1. A group sets up Container A (moist cotton wool) and Container B (dry cotton wool) to test whether maize and bean seeds need water to germinate. Which of these must be kept the SAME in both containers for the test to be fair?

- A. The type and number of seeds and the temperature**
- B. Whether the seeds are alive or boiled
- C. The amount of water given
- D. Whether the cotton wool is wet or dry

Correct answer: A. In a fair test only one variable (water) is changed. The seed type, number of seeds and temperature must stay the same so any difference is caused by water alone.

USE AFTER: OBSERVE PHASE — Part A: Set E, boiled-then-cooled seeds on moist cotton

Q2. In the germination investigation, one container holds boiled and then cooled maize and bean seeds on moist cotton wool with air and room temperature. Why is this container included?

- A. To show that a seed must be alive (viable) to germinate even when water, air and warmth are present**
- B. To show that seeds need cold conditions to germinate
- C. To show that seeds germinate faster when they are boiled
- D. To show that seeds need more water than the control container

Correct answer: A. Boiling kills the embryo. If these seeds fail in good conditions, it shows a viable (living) seed is needed as well as water, oxygen and warmth.

USE AFTER: OBSERVE PHASE — Part B: examining germinating maize and bean seedlings and drawing hypogeal and epigeal germination

Q3. A learner examines two seedlings. In seedling X, the coleoptile and plumule push up while the cotyledon stays below the soil. In seedling Y, the hypocotyl arches and lifts the cotyledons above the soil. Which statement is correct?

- A. Both X and Y are epigeal
- B. X is bean (epigeal) and Y is maize (hypogeal)
- C. Both X and Y are hypogeal
- D. X is maize (hypogeal) and Y is bean (epigeal)**

Correct answer: D. Maize shows hypogeal germination, where the cotyledon stays below the soil. Bean shows epigeal germination, where the hypocotyl lifts the cotyledons above the soil.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning: Container C (submerged seeds with oil on top)

Q4. Maize seeds are fully submerged in boiled, cooled water with a thin layer of cooking oil on top. After 5 days, very few have germinated, even though there is plenty of water. What is the best explanation?

- A. The water was too warm for enzymes to work
- B. The seeds had no soil to grow in
- C. The seeds received too much light
- D. Too little oxygen is available for aerobic respiration, so the seeds cannot release the energy needed for germination**

Correct answer: D. The oil and boiled water keep oxygen out. Without oxygen, aerobic respiration cannot release energy for germination. This also explains why seeds in waterlogged soil may fail.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning using results from Container A (control)

Q5. In a germination test, 9 of 10 bean seeds germinated on moist cotton wool at room temperature with no soil. Which conclusion is supported by this evidence?

- A. Seeds need fertiliser to germinate
- B. Seeds need soil to germinate
- C. Seeds germinate only in the dark
- D. Soil is not necessary for germination if water, oxygen and suitable warmth are present**

Correct answer: D. Seeds germinated without any soil, so soil is not a requirement. Water, oxygen and suitable temperature were enough, which corrects the common misconception that soil is essential.

USE AFTER: MODEL BUILDING PHASE — Revising the Day 1 model: conditions and labelled diagrams of maize and bean germination

Q6. Which list shows the correct order of events in the germination of a maize or bean seed?

- A. Water uptake, enzymes activated, radicle emerges, plumule emerges**
- B. Radicle emerges, water uptake, enzymes activated, plumule emerges
- C. Enzymes activated, plumule emerges, water uptake, radicle emerges
- D. Plumule emerges, radicle emerges, water uptake, enzymes activated

Correct answer: A. The seed first takes in water, which activates enzymes. The radicle then breaks out and grows downwards, followed by the plumule.

USE AFTER: END OF LESSON — Model revision note: why the farmer's seeds planted together after the same rain germinate at different times or fail

Q7. A farmer in Kitale plants healthy-looking maize seeds after good rain. The soil is moist, well aerated and warm, yet some seeds never germinate. Which is the most likely reason?

- A. The seeds are hypogeal, so they cannot germinate
- B. The seeds did not get enough sunlight underground
- C. The seeds are dead (not viable) or dormant**
- D. The soil had too much oxygen

Correct answer: C. When water, oxygen and warmth are suitable, failure points to an internal factor: the seed is dead (non-viable) or still dormant. Seeds do not need light to germinate, and oxygen is needed, not harmful.